Asteroid Ocean Impact

Team 300 : Problem B

November 7, 2021

1 Abstract

An asteroid named "Urraal-Pongkhi" has hit in the ocean, 1,000 km away from a coastal city. So here we have found the minimum mass of that asteroid to cause substantial damage to this coastal city. Firstly we have considered asteroid impact threats, airburst and energy disposition to understand how much energy is yielded from an asteroid with a certain amount of mass. Then we checked the shock wave speed. After that, we have studied dispersive tsunami propagation to find the energy related to this. From that information, we have found the minimum mass of that asteroid.

2 Asteroid Impact Threat

Asteroid impacts pose a continual threat to humanity. For example, on February 15, 2013, an asteroid penetrated the atmosphere over Chelyabinsk, Russia, entering at a horizon angle of approximately 18°, and releasing energy equivalent to 570 ± 150kt TNT. For comparison, the nuclear weapon that was detonated approximately 0.5km above the ground in Hiroshima, Japan yielded approximately 12.5kt TNT. The main airburst over Chelyabinsk occurred at an approximate altitude of 30km and created a shock wave strong enough to shatter windows out to a distance of 120km from the asteroid track, injuring over 1,200 people in Chelyabinsk city and hundreds more in nearby towns and rural areas. Had the asteroid approached from a higher angle, more serious damage would have occurred from higher concentration of the impact blast energy on the ground. Sixteen hours after the meteorite struck near Chelyabinsk, the 45m diameter asteroid 2012 DA14 approached to within 27,743km of Earth's surface—inside the orbit of geosynchronous satellites. If DA14 were to strike Earth, it would deliver approximately 7.2Mt TNT. Although the Chelyabinsk meteorite and DA14 arrived at or near Earth on the same day, the two objects were not linked to each other, coming from completely unrelated orbits. That two such seemingly improbable events could occur within hours of each other serves as a stark reminder that humanity is continually at risk of asteroid impact. Asteroids at least the size of DA14 (~50m diam.) are expected to strike Earth approximately every 650 years, while objects at least the size of the Chelyabinsk asteroid (~20m diam.) are expected to strike Earth approximately every 50-100 years. For example, the Tunguska 1908 event could have caused large scale loss of life, but did not due to the remote area in Russia it airburst over. This event is estimated to have been a roughly 65m diameter asteroid (or possibly an atmospheric grazing comet) with an air blast energy yield of between 3 and 30Mt. This event flattened more than 2000km2 of forest as seen from the 1927 Soviet Academy of Sciences expedition. Larger objects also pose a severe threat, as the total

kinetic energy associated with an impact of a 100m asteroid is equivalent to approximately 100Mt TNT, and that of the well-known ~350m threat, Apophis, would have an impact yield of approximately 3-4Gt TNT, or about ½ of the Earth's total nuclear arsenal, while Bennu at 490m could have a yield greater than the entire Earth's nuclear arsenal. For reference, Apophis will next visit Earth on Friday April 13, 2029 and come within the Figure 1 – Exo-atmospheric kinetic energy (KE) vs diameter and atmospheric entry speed. Nuclear weapon yields and total human nuclear arsenal shown for comparison. 6 geosync belt. Effective mitigation strategies are imperative to ensure humanity's continuity and future advancement. Every day, approximately 100 tons of small debris impact the Earth and the effect is virtually undetectable. If this consisted instead of a single 100-ton asteroid with an effective diameter of about 4 meters, the effect would be quite noticeable, but not a serious threat. This is the basic idea behind our approach: reduce the threat to a large number of small objects that primarily burn up or airburst in the atmosphere, and even if small remnants impact, as in a short response threat scenario for high density asteroids, they will produce very little damage compared to the same mass in the parent asteroid. Extremely large asteroids (>1km diameter) and comets pose unique threats that we briefly discuss as a part of this program. Existential threats to humanity are very low recurrence, but are known to have happened multiple times in the past including the last mass extinction approximately 65Myr ago when an estimated 10km diameter asteroid triggered large scale life extinction, including the destruction of dinosaurs. The equivalent yield of this event is estimated to have been approximately 100 Tera-tons TNT, or about 15,000 times larger than the current nuclear arsenal.

3 Airburst and Energy Deposition

Stony asteroids with diameters smaller than about 85m will generally airburst in the atmosphere depending on their composition, speed, and angle

of impact. While this may seem to alleviate the problem as the bulk of the asteroid never reaches the Earth's surface, the problem is that the airburst altitude decreases with increased size and the damage from the airburst shock waves can be extremely dangerous, as was the case in the Tunguska (~100m diameter; estimates are 50-190m and 3- 30MT yield, 5-10km airburst altitude - 1908) and Chelyabinsk (~20m diameter, 19km/s, 500Kt yield, 30km airburst altitude - 2013) events.

4 Physics of Air Bursts and Blast Waves

The physics of chemical and nuclear detonations and the generation of the resultant blast waves has been well studied theoretically, as well as measured experimentally. The relevant detonation energy levels for the relevant cases of asteroids impacting the Earth are much closer to the yields of nuclear weapons than to the chemical explosives. As shown above, for the case of asteroids in the 1-100m diameter class moving at relevant speeds of 15-50km/s, the energies involved are in the low kt to high Mt range, or much above common chemical explosive devices. The relevant energy ranges span the regime from small tactical to the largest strategic nuclear devices ever tested. Some of the early solutions to blast waves are discussed in the 1947 LANL paper of Bethe et al [20], as well as in papers by Sedov, von Neuman, and Taylor [20]–[23], and by Zeldovich in the USSR [24]. They all dealt with the energy singularity problem relevant to compact explosive devices such as nuclear weapons. Some of the experimental data is summarized in the 1977 Effects of Nuclear Weapons by Glasstone and Dolan [2], wherein a 1kt "standard" is given which allows scaling to arbitrary yields. While much of the nuclear weapons test data is primarily focused on "close in" blast wave effects, we can scale to larger distances. While at distances close to the detonation, the shock wave speed is extremely supersonic. Then shock rapidly decays to near sound speed as we discuss below. Most of the relevant cases of interest in this paper are for long range effects where the peak over pressure is expected to scale as 1/r if there were no atmospheric absorption. We discuss absorption in the atmosphere below. As shown

earlier, the typical fraction of nuclear weapon yield that goes into the blast wave for air burst detonations is about 50%. The formation of the shock wave is essentially adiabatic due to the extremely short time scales relative to the thermal diffusion time scales. The shockwave from an asteroid air burst starts at extremely high speeds (at the speed of the bursting asteroid/fragment, which is typically > Mach 30 Figure 8 – Sea level nuclear air burst "1kt standard" with scaling to 1Mt nuclear yield. Pressure vs distance from the point of detonation. Power law fits are shown with scaling to arbitrary yield and distance. For asteroids, we are not generally concerned with ranges less than 30km. A 10m diameter asteroid (20km/s, 2.6g/cc, 45deg angle) with 48kt blast yield is shown as a 100kt nuclear equivalent assuming approx. 50% nuclear yield to blast yield. 14 at production), but then, like the case of a nuclear weapon, it rapidly decreases with increasing distance and decays to sound speed (~330m/s). We model the peak over pressure from the blast wave as two power laws covering the "near" and "far" distance regimes.

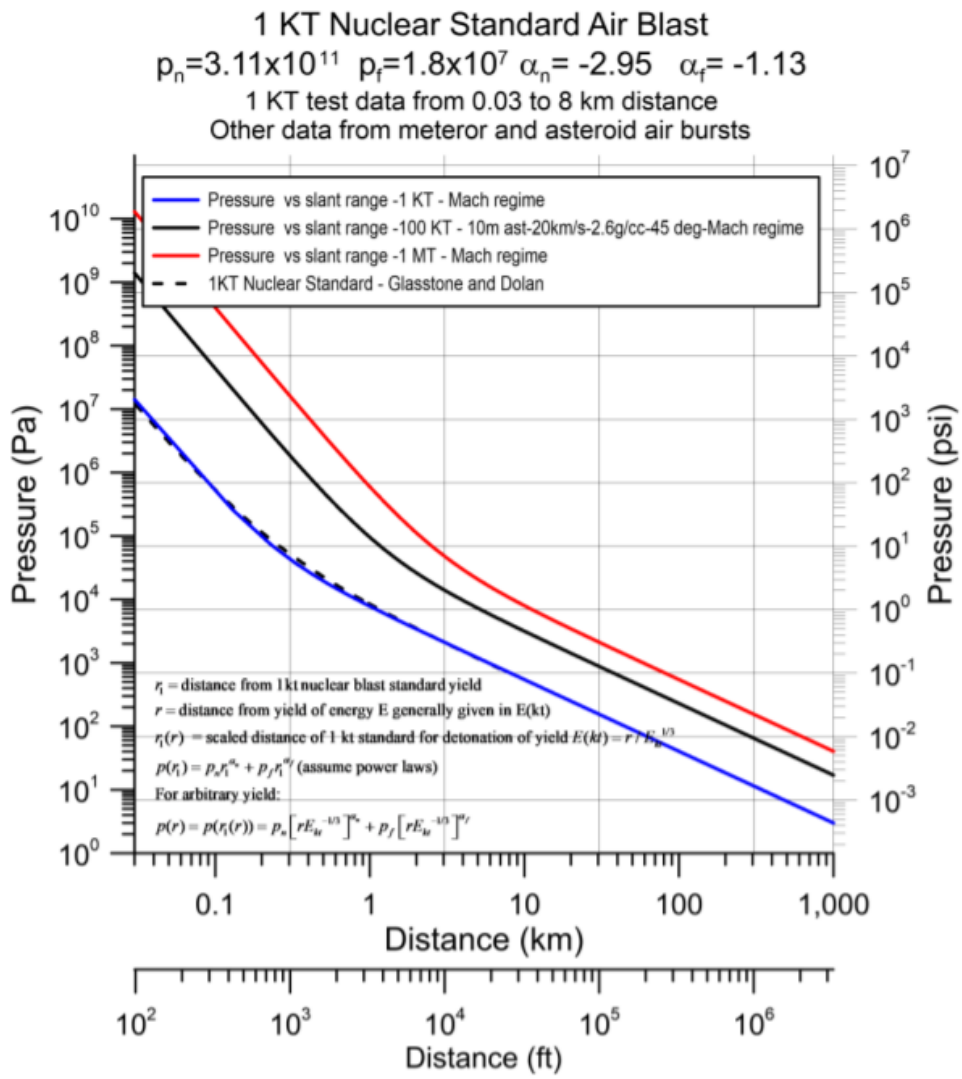

Fig.1 – Sea level nuclear air burst "1kt standard" with scaling to 1Mt nuclear yield. Pressure vs distance from the point of detonation. Power law fits are shown with scaling to arbitrary yield and distance. For asteroids, we are not generally concerned with ranges less than 30km. A 10m diameter asteroid (20km/s, 2.6g/cc, 45deg angle) with 48kt blast yield is shown as a 100kt nuclear equivalent assuming approx. 50% nuclear yield to blast yield.

r_1(m) = distance from 1kt nuclear blast standard yield

r(m) = distance from yield of energy E given in E(kt)

$r_1(r)$ = scaled distance of 1 kt standard for detonation of yield E(kt) $=r/E_{kt}^{1/3}$

$p(r_1) = p_n r_1^{\alpha_n} + p_f r_1^{\alpha_f}$ (assume power laws)

For arbitrary yield:

$P(r) = p(r_1(r)) = p_n[rE_{kt}^{-1/3}]^{\alpha_n} + p_f[rE_{kt}^{-1/3}]^{\alpha_f}$

A good fit to the measured data from atmospheric nuclear tests is:

$p_n = 3.11 \times 10^{11}$ Pa

$\alpha_n = -2.95$

$p_f = 1.8 \times 10^{7}$ Pa

$\alpha_f = -1.13$

Transition distance between near and far distances in power law:

$r_{tr} = E_{kt}^{-1/3} [p_n / p_f]^{1/(\alpha_f - \alpha_n)}$

5 Shock wave speed

The speed of the shock wave generated in an air burst depends on the peak pressure of the blast. The shock process is largely adiabatic since the time scale for thermal mixing is much longer than the relevant acoustical time scales. We calculate the speed vs pressure ratio, as well as vs distance. The latter is based on the peak pressure calculated above from nuclear weapons tests. Combining the measured weapons test data with the theoretical shock speed vs pressure ratio allows the calculation of shock speed vs distance. In general, we see the shock speed decays from highly supersonic to the speed of sound relatively quickly.

$v_{shock} / v_{sound} = (1+ 6P/7P_0)^{1/2}$

P = shock pressure

P_0 = ambient air pressure

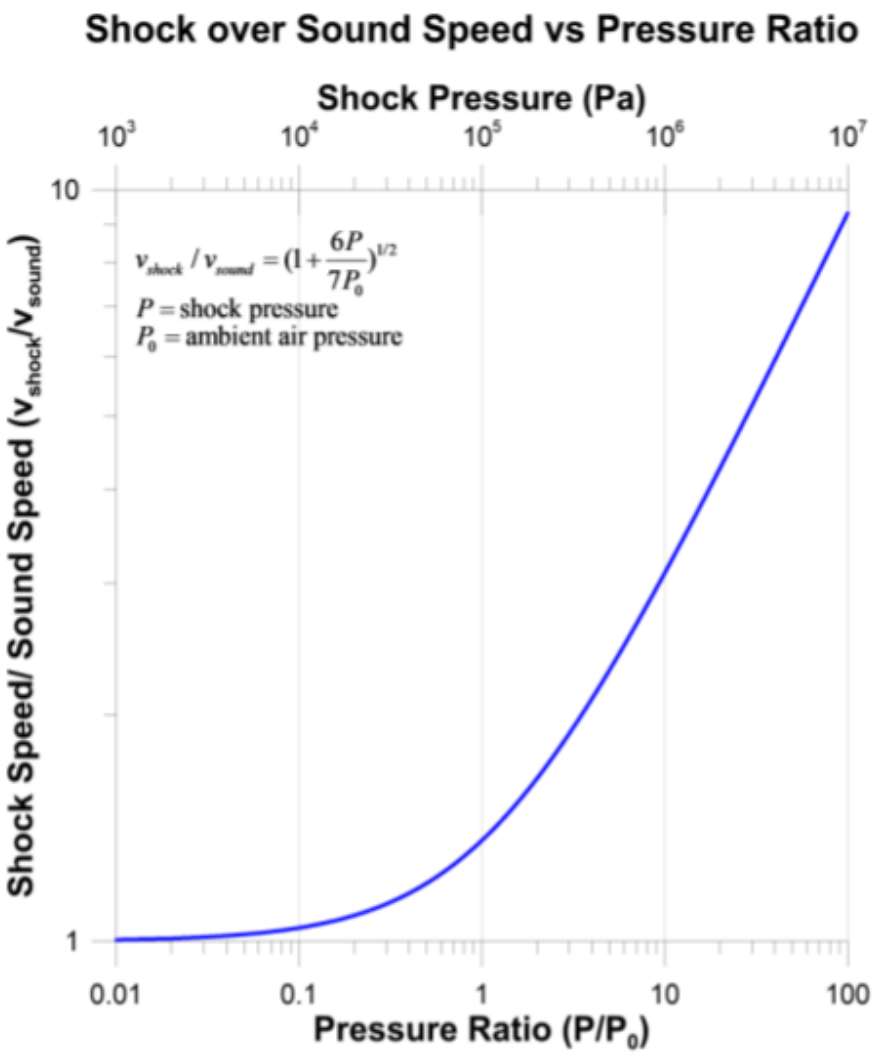

Fig.2 : The Mach #, or ratio of shock speed over sound speed, as a function of the peak pressure P over ambient pressure Po. At low pressure ratios, the shock speed is close to the sound speed. At high pressure ratios, the shock wave is highly supersonic. However, typically the blast wave peak pressure is << Po for the fragment size and distance to the observer in cases of interest here. This results in the shock speed being close to sound speed.

6 Pressure – Flux and Sound Pressure Level & Hearing Damage

The relationship between the blast wave pressure P (Pa) and the acoustical flux I (W/m2) as well as what is referred to as the Sound Pressure Level (SPL) will caused due to that. Shock pressure will also damage the hearing of people. The threshold for hearing damage varies with the frequency and pulse shape. Typically, even short exposures to an SPL of >150db can cause hearing loss. An SPL of 150db corresponds to a peak pressure of approximately 1kPa (lower limit of window breakage), which corresponds to an acoustical peak flux of about 10kW/m2 .

7 Existential Threats

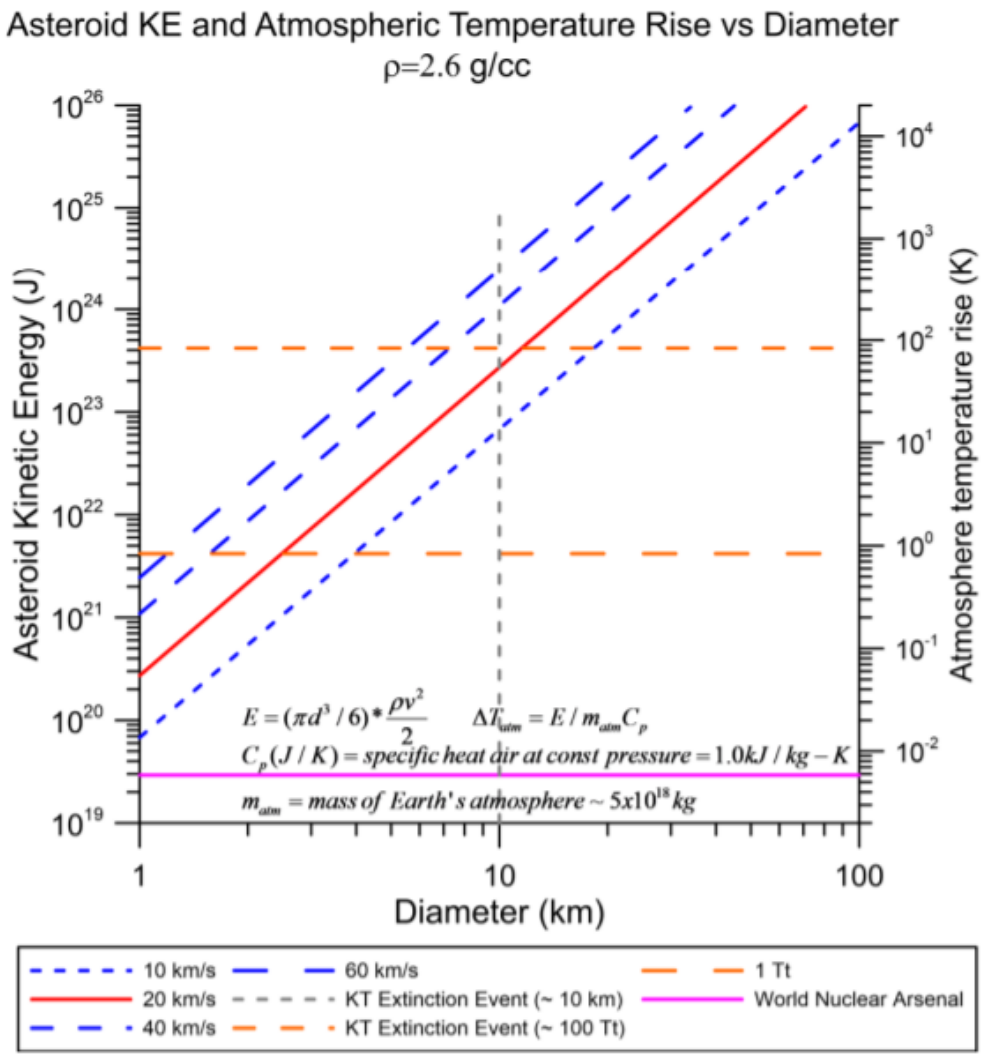

Fig.3: Asteroid exo-atmospheric KE vs diameter from 1 to 100 km (extremely large asteroids) and speed for density 2.6 g/cc. The right hand y-axis shows the temperature rise of the Earth's atmosphere assuming equilibrium. To first order, this plot is independent of fragmentation. Above a few km diameters, the effects on the Earth's atmosphere become serious, and critical above 10km. In these cases, either the interception for fragmentation must be done early enough to spread the debris cloud to be large enough so that most of the fragments miss the Earth, or complete diversion of the asteroid to miss the Earth must be implemented.

Fortunately, very large asteroids are both extremely rare (though they have hit the Earth in our geologic history) and we would have sufficient notice to take action long before impact. The KT event we believe that caused the mass extinction 65 MYr ago is shown at 10km diameter based on the Iridium in the KT boundary clay layer.

8 Dispersive Tsunami Propagation

The Shallow Water and Boussinesq Equations

The shallow water equations can be written

$$h_t + (hu)_x = 0$$

$$(hu)_t + \left(hu^2 + \frac{1}{2} gh^2 \right)_x = -ghB_x$$

where $h(x, t)$ = water depth,

$B(x)$ = topography ($B < 0$ offshore),

$\eta(x, t) = B(x) + h(x, t)$, with $\eta = 0$ being sea level.

Thus $h_0(x) = -B(x)$ is the depth of water at rest. The depthaveraged horizontal velocity is $u(x, t)$, and so hu is the momentum, and finally $g = 9.81$ is the gravitional constant. These equations are a long wave length approximation to the Euler equations, in the limit of small ocean depth relative to the wavelength L of the disturbance. For earthquake generated tsunamis, a typical ocean depth might be 4 km, and a subduction zone can have a wavelength of 100 km or more, giving a small ratio. The equations (3.1) have the form of a hyperbolic system of conservation laws with a source term in the momentum equation that is nonzero only on varying topography. The GeoClaw implementation incorporates the topography term into the Riemann solvers in order to obtain a well-balanced method [16], which amounts to solving the equations (3.1) in the nonconservative form

$$h_t + (hu)_x = 0$$

$(hu)_t + hu2\ \ x + gh\eta x = 0.$

Peregrine derived a Boussinesq-type extension on a flat bottom in the form

$ht + (hu)x = 0$

$(hu)t + hu2\ \ x + gh\eta x - 1/3\ h^2{}_0\ (hu)_{txx} = 0$

These equations have some drawbacks however, and do not match the dispersion relation of the Euler equations as well as other models developed more recently. (For a historical review of Boussinesq-type models. Madsen and Sorenson and Shaffer and Madsen [26] optimized the equations by adding a term with a parameter B1 that could be chosen to match the water wave dispersion relation more closely. On general topography, they obtained

$$\begin{aligned} h_t + (hu)_x &= 0 \\ (hu)_t + \left(hu^2\right)_x + gh\eta_x &= \left(B_1 + \frac{1}{2}\right) h_0^2 (hu)_{txx} \\ &\quad + \frac{1}{6} h_0^3 (hu/h_0)_{txx} - B_1 h_0^2 g (h_0 \eta_x)_{xx}, \end{aligned}$$

$$\begin{aligned} h_t + (hu)_x &= 0 \\ (hu)_t + \left(hu^2\right)_x + gh\eta_x - D_{11}((hu)_t) &= gB_1 h_0^2 (h_0 \eta_x)_{xx}. \end{aligned}$$

$$[I - D_{11}]\psi = -D_{11}\left[\left(hu^2\right)_x + gh\eta_x\right] + gh_0^2 B_1 (h_0 \eta_x)_{xx}.$$